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भाग 2 विशेष अपेक्षाएँ

अनुभाग 32 मालिश उपकरण

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 32 Massage Appliances

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

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Price Group 3

Electrical Appliances Sectional Committee, ETD 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This standard covers general and safety requirements of electrical appliances intended for massage purposes. This standard does not cover performance requirements which is covered under a separate composite standard IS 7137 : 1973.

This standard is to be read in conjunction with IS 302-1 (1979). For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979). Instead of reproducing full text of each clause, clauses of IS 302-1 (1979) which are applicable (which means that relevant provisions of the clause apply) or not applicable and the sub-clauses or portions thereof which are not applicable are indicated as under:

- a) In case of a clause where it is applicable or not applicable, the wording used is 'This clause of IS 302-1 (1979) is applicable/not applicable'.
- b) In case of sub-clause or part thereof 'Not applicable' wherever a sub-clause of IS 302-1 (1979) is to be replaced by a new text, it has been indicated as under:

'Replacement — followed by the new text'.

Any addition to the existing provisions of a sub-clause of IS 302-1 (1979) has been indicated as under:

'Addition—followed by the text of the additional matter'.

Clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101 and additional sub-clauses are numbered with the main clause number followed by 101, 102 etc, for example, 4.101.

Should however any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from the following publications issued by the International Electrotechnical Commission:

IEC Pub 335-1 (1976) Safety of household and similar electrical appliances: Part 1 General requirement.

IEC Pub 335-2-32 (1987) Safety of household and similar electrical appliances: Part 2 Particular requirements for massage appliances.

The Indian Standards listed in Annex AA are necessary adjunct to this standard.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 32 Massage Appliances

1 SCOPE

This clause of IS 302-1(1979) is applicable except as follows :

1.1 Replacement

This standard applies to electrical appliances primarily intended for massage purposes and for use under household and similar conditions, their rated voltage not exceeding 250 V, ac single phase or dc.

Appliances not intended for normal household use but which nevertheless may be a source of danger to the public, such as appliances intended to be used by lay men in places other than the home are within the scope of this standard.

Appliances intended primarily for purposes other than massage, but which have accessories for massage purposes are within the scope of the standard covering the primary function.

This standard does not take into account the special hazards which exist in nurseries and other places where there are young children or aged or infirm persons without supervision; in such cases additional requirements may be necessary.

2 TERMINOLOGY

This clause of IS 302-1(1979) is applicable except as follows :

2.30 Replacement

Normal load denotes the load obtained when the appliance is operated under the following conditions. Hand held massage appliances are operated with no external load other than that imposed by the vibrator head.

Massage appliances intended to be used under the foot or feet of a sitting person are operated with the platform loaded with a mass of 5 kg evenly distributed over an area of 10 cm × 30 cm.

Massage appliances intended to be used under the feet of a standing person are operated with the platform load with a mass of 80 kg evenly distributed over an area of 10 cm × 30 cm.

Massage belts are operated with the belt loaded by means of a roller with a force of 200 N applied in a horizontal direction so that the belt is approximately symmetrical about the roller.

Massage pads intended to be used under a sitting person are operated with the pad placed between two

layers of felt, this assembly being supported by a flat horizontal surface and the upper layer of felt being loaded with a mass of 80 kg, evenly distributed over an area of 30 cm × 30 cm. Each layer of felt has a thickness of 25 mm and a mass of 4 ± 0.4 kg/m², both layers extending at least 15 mm beyond the edges of the pad.

Massage chairs are operated with no load.

Massage beds intended to support a lying person are operated with the surface on which the person is intended to lie loaded with a mass of 80 kg evenly distributed over an area of 50 cm × 200 cm.

Additional Sub-clause :

2.101 Electric Massager — A hand held, portable electrical appliances for simulating the muscles of the human body. It includes cables and connectors.

3 GENERAL REQUIREMENT

This clause of IS 302-1(1979) is applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1(1979) is applicable except as follow :

Additional Sub-clause :

4.101 Massage appliance are tested as portable appliances, unless they are designed to be fixed to a support in normal use.

Hand-held appliances are subjected to the additional tests specified for such appliances.

5 RATING

This clause of IS 302-1(1979) is applicable.

6 CLASSIFICATION

This clause of IS 302-1(1979) is applicable.

7 MARKING

This clause of IS 302-1(1979) is applicable.

Additional Sub-clause :

7.101 The massage appliances may also be marked with the standard mark.

7.102 The use of the standard mark is governed by the provisions of Bureau of Indian Standards Act,

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1986 and the rules and regulations made thereunder. The details of conditions under which the licence for the use of the standard mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1(1979) is applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of IS 302-1(1979) is applicable.

10 INPUT AND CURRENT

This clause of IS 302-1(1979) is applicable.

11 TEMPERATURE-RISE

This clause of IS 302-1(1979) is applicable except as follows :

11.2 Addition

Massage appliances incorporating heating elements are also positioned as specified for motor-operated appliances.

11.7 Replacement

Hand-held massage appliances are operated for 20 min. Other massage appliances are operated continuously until steady conditions are established.

11.8 Addition

The temperature rise of parts which in normal use are in contact with skin or hair or are held in the hand shall not exceed the limits specified for handles which in normal use are continuously held. The temperature rise of such parts is determined for every accessory intended to be used with the appliance.

For massage pads with heating elements the temperature limits specified for heating pads in the relevant Indian Standard for heating pads shall apply.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

This clause of IS 302-1(1979) is applicable.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

This clause of IS 302-1(1979) is applicable.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1(1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1(1979) is applicable.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (After humidity treatment)

This clause of IS 302-1(1979) is applicable.

17 OVERLOAD PROTECTION

This clause of IS 302-1(1979) is applicable.

18 ENDURANCE (Under Consideration).

19 ABNORMAL OPERATION

This clause of IS 302-1(1979) is applicable except as follows :

19.1 Replacement

Instead of the indication of the sub-clauses applicable to the various types of appliances the following applies:

Compliance is checked by the tests of the following sub-clauses :

- for Class II massage appliances with heating elements and for massage beds with heating elements, if these appliances are provided with a control which limits the temperature during the test of Clause 11.....Sub-clause 19.4.
- for hand-held massage appliances with heating elements other than appliances which have to be kept switched on by hand..... Sub-clause 19.5.
- for massage appliances incorporating motors having a locked rotor torque smaller than the full load torque for those having a capacitor not complying with IS 2993 : 1975 ac motor capacitors, in the circuit of an auxiliary winding and for massage beds..... Sub-clause 19.6.
- for massage beds, other than those incorporating magnetically-driven vibrators Sub-clause 19.8.
- for massage appliances incorporating series motors Sub-clause 19.10.

19.4 Replacement

The appliance is operated under normal load, any thermal control which limits the temperature during the test of Clause 11 being short-circuited. Heating elements are operated at a supply voltage such that the input to these elements is 1.24 times their maximum rated input, and the motor is operated at a voltage equal to 1.06 times the rated voltage of the appliance.

If the appliance is provided with more than one control these are short-circuited in turn.

19.5 Replacement

The test of Sub-clause 19.4 is repeated but with the thermal control not short-circuited. The test is continued until steady conditions are established irrespective of the rated operating time.

19.6 Replacement

Instead of the text preceding the table the following applies :

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The appliance starting from cold is switched on with the moving parts locked and at a supply voltage equal to rated voltage or to the upper limit of the rated voltage range. The appliance is initially loaded as specified for normal load for a period of 30 s after which the load is removed and the test continued until steady conditions are established or if a timer is provided for a period equal to the maximum period allowed by the timer.

If the motor has a capacitor not complying with IS 2993 : 1975 in the circuit of an auxiliary winding the capacitor is short-circuited or open-circuited whichever is the more unfavourable. If there are two or more capacitors the capacitors are short-circuited or open-circuited one at a time.

At the end of the test period specified or at the instant of operation of fuses, thermal cut-outs, motor protection devices and the like the temperature of the windings shall not exceed the values shown in the following table :

If the appliance incorporates more than one motor the test is made for each motor separately.

Alternative tests for protected motor units are given in Annex D.

19.10 Addition

Detachable parts which may influence the load are removed.

20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1(1979) is applicable except as follows :

20.1 Addition

During the test massage chairs are loaded.

Additional Sub-clause :

20.101 Massage appliances shall be so designed that in normal use hair cannot be drawn into the appliance or be entangled in moving parts.

Compliance is checked by inspection.

21 MECHANICAL STRENGTH

This clause of IS 302-1(1979) is applicable except as follows :

21.1 Addition

The foot platform of massage appliances intended to be used under the foot of a sitting person is then loaded for 30 s with a mass of 80 kg evenly distributed over an area of 10 cm × 30 cm.

22 CONSTRUCTION

This clause of IS 302-1(1979) is applicable except as follows :

22.1 Replacement

Portable massage appliances shall be of Class II or Class III.

Stationary massage appliances shall be of Class I, Class II or Class III.

Compliance is checked by inspection and by the relevant test.

23 INTERNAL WIRING

This clause of IS 302-1(1979) is applicable.

24 COMPONENTS

This clause of IS 302-1(1979) is applicable except as follows :

24.1 Addition

Switches incorporated in massage appliances are not required to be switches for frequent operation.

24.2 Replacement

Instead of the text of a) the following applies :

— Switches in flexible cables or cords, except for massage pads :

24.9 Replacement

Massage pads shall be fitted with a switch in their supply circuit.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORDS

This clause of IS 302-1(1979) is applicable except as follows :

25.1 Addition

According to IS 3010 (Parts 1 and 2) : 1965 cord sets incorporating a flat twin tinsel cord are only allowed for appliances provided with a miniature appliance inlet.

25.6 Addition

Power supply cords massage appliances shall be not lighter than ordinary tough rubber sheathed flexible cord as per IS 9968 (Part 1) : 1988 or pvc insulated flexible cords as per IS 694 : 1990.

Power supply cords of massage appliances shall have a length of at least 2.0 m, if supplied with a plug, or it shall be not less than 2.05 m if supplied without plug.

Type Z attachment is allowed.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1(1979) is applicable.

27 PROVISION FOR EARTHING

This clause of IS 302-1(1979) is applicable.

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28 SCREWS AND CONNECTIONS

This clause of IS 302-1(1979) is applicable.

29 CREEPAGE DISTANCES & CLEARANCES

This clause of IS 302-1(1979) is applicable.

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This clause of IS 302-1(1979) is applicable.

31 RESISTANCE TO RUSTING

This clause of IS 302-1(1979) is applicable except as follows :

31.1 Replacement

The ferrous part, the rusting of which might cause the appliance to fail to comply with this standard, shall be adequately protected against rusting.

32 RADIATION HAZARDS

This clause of IS 302-1(1979) is not applicable.

101 TESTS

101.1 Type Tests

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of massage appliances of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the massage appliances shall be visually examined and inspected for obvious visual defects in respect of components, parts and their assembly, construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria for Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority, at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

Table 101 Schedule of Type Tests

(Clause 101.1)

Sl No.	Test	Clause Reference
i)	Protection against electric shock	8
ii)	Starting of motor operated appliances	9
iii)	Input and current	10
iv)	Temperature rise	11
v)	Operation under overload conditions	12
vi)	Electrical insulation and leakage current at operating temperature	13

Table 101 Schedule of Type Tests (Continued)

(Clause 101.1)

Sl No.	Test	Clause Reference
vii)	Radio and television interference suppression	14
viii)	Moisture resistance	15
ix)	Insulation resistance and electric strength (after humidity treatment)	16
x)	Overload protection	17
xi)	Endurance (under consideration)	18
xii)	Abnormal operation	19
xiii)	Stability and mechanical hazards	20
xiv)	Mechanical strength	21
xv)	Construction	22
xvi)	Internal wiring	23
xvii)	Components	24
xviii)	Supply connections and external flexible cables and cords	25
xix)	Terminals for external conductors	26
xx)	Provision for earthing	27
xxi)	Screws and connections	28
xxii)	Creepage distance and clearances	29
xxiii)	Resistance to heat fire and tracking	30
xxiv)	Resistance to rusting	31
xxv)	Radiation hazards	32

101.2 Acceptance Tests

The following shall constitute the acceptance tests :

Sl No.	Test	Clause Reference
a)	Protection against electric shock	8
b)	Input	10
c)	Temperature rise	11
d)	Electrical insulation and leakage current at operating temperature	13
e)	Moisture resistance	15
f)	Insulation resistance and electric strength (after humidity treatment)	16
g)	Provision for earthing	27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1(1979).

101.3 Routine Test — The following shall constitute the routine tests :

Sl No.	Test	Clause Reference
a)	Protection against electric shock	8
b)	High voltage	13.3.2 of IS 302-1(1979)
c)	Provision for earthing	27

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ANNEX A
TABLES OF TYPE TEST

Appendix A of IS 302-1(1979) is not applicable.

ANNEX B
(*Clause 101.2.1*)

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

Appendix B of IS 302-1(1979) is applicable.

ANNEX C
ELECTRONIC CIRCUITS

Appendix C of IS 302-1(1979) is applicable if electronic circuits are used.

ANNEX D
MEASUREMENT OF TEMPERATURE WITH THERMOMETERS

Appendix D of IS 302-1(1979) is applicable.

ANNEX E
ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of 302-1(1979) is applicable.

ANNEX F
IMPACT TEST APPARATUS

Appendix F of IS 302-1(1979) is applicable.

ANNEX G
THERMAL CONTROLS AND OVERLOAD RELEASES

Appendix G of IS 302-1(1979) is applicable.

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ANNEX H

MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302-1(1979) is applicable.

ANNEX J

TEST FOR FIRE RESISTING PROPERTIES

Appendix J of IS 302-1(1979) is applicable.

ANNEX K

BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER AND NICKEL PLATING

Appendix K of IS 302-1(1979) is not applicable.

ANNEX L

APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON NICKEL, STEEL AND COPPER

Appendix L of 302-1(1979) is not applicable.

ANNEX AA

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
302-1(1979)	Safety of household and similar electrical appliances : Part 1 General requirement		three-pin type) : Part 1 Appliance connectors
694 : 1990	PVC insulated cable for working voltages up to and including 110 V (<i>second revision</i>)	3010 (Part 2) : 1965	Appliance connectors and appliance inlets (non-reversible three-pin type) : Part 2 Appliance inlet
2993 : 1975	Motor capacitors (<i>first revision</i>)	9968 (Part 1) : 1988	Elastomer insulated cables : Part 1 For working voltage up to and including 1 100 V (<i>first revision</i>)
3010 (Part 1) : 1965	Appliance connectors and appliance-inlets (non-reversible		

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BUREAU OF INDIAN STANDARDS

Headquarters:

Manak Bhavan, 9 Bahadur Shah Zafar Marg, New Delhi 110002
Telephones : 331 01 31, 331 13 75

Telegrams : Manaksanstha
(Common to all Offices)

Regional Offices:

Telephone

Central : Manak Bhavan, 9 Bahadur Shah Zafar Marg
NEW DELHI 110002

{ 331 01 31
331 13 75

Eastern : 1/14 C. I. T. Scheme VII M, V. I. P. Road, Maniktola
CALCUTTA 700054

{ 37 84 99, 37 85 61
37 86 26, 37 86 62

Northern : SCO 445-446, Sector 35-C, CHANDIGARH 160036

{ 53 38 43, 53 16 40
53 23 84

Southern : C. I. T. Campus, IV Cross Road, MADRAS 600113

{ 235 02 16, 235 04 42
235 15 19, 235 23 15

Western : Manakalaya, E9 MIDC, Marol, Andheri (East)
BOMBAY 400093

{ 632 92 95, 632 78 58
632 78 91, 632 78 92

Branches : AHMADABAD. BANGALORE. BHOPAL. BHUBANESHWAR. COIMBATORE.
FARIDABAD. GHAZIABAD. GUWAHATI. HYDERABAD. JAIPUR. KANPUR.
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